

DualCAM++: Dual-saliency Guided Sample Selection for Plug-and-Play Backdoor Enhancement

1st Chuangliang Zhang

School of Informatics

Xiamen University

Xiamen, China

31520231154325@stu.xmu.edu.cn

2nd Qingyuan Zeng

School of Informatics

Xiamen University

Xiamen, China

36920221153145@stu.xmu.edu.cn

3rd Yunpeng Gong

School of Informatics

Xiamen University

Xiamen, China

fmonkey625@gmail.com

4th Zhenzhong Wang*

School of Informatics

Xiamen University

Xiamen, China

zhenzhong16.wang@connect.polyu.hk

5th Jiajing Lin

School of Informatics

Xiamen University

Xiamen, China

31520231154298@stu.xmu.edu.cn

6th Min Jiang

School of Informatics

Xiamen University

Xiamen, China

minjiang@xmu.edu.cn

Abstract—Backdoor attacks inject a few trigger-label pairs into the training data, causing deep-learning models to behave normally on clean inputs yet consistently misclassify any input stamped with the trigger as an attacker-specified target. Existing methods perform well at moderate poisoning rates, but their effectiveness drops sharply in the ultra-low regime ($\rho \leq 1\%$) required for real-world stealth. We propose DualCAM++, a plug-and-play sample-selection module that fuses two complementary Grad-CAM saliency cues—*Heat-Sum* (global activation) and *Heat-Max* (local peak)—via a Borda count, selecting only the most vulnerable images for poisoning without changing the trigger pattern or training pipeline. Experiments demonstrate that DualCAM++ markedly boosts attack success while preserving clean accuracy under ultra-low poisoning rates and integrates seamlessly with a variety of backdoor frameworks. Ablation studies confirm that neither saliency cue alone suffices; their fusion is essential for maximal benefit. Notably, at a 0.5% poisoning rate in the Blended attack, our module improves the attack success rate by more than 6%.

Index Terms—backdoor attack, data poisoning, saliency map, Grad-CAM.

I. INTRODUCTION

In recent years, backdoor attacks have emerged as a central topic in the security of deep learning. When an adversary stealthily implants specific “trigger-target label” mappings into the training set, the resulting model performs normally on clean inputs but produces attacker-specified outputs when presented with trigger-containing inputs. Classical backdoor methods such as BadNets [10] and Blended [11] demonstrate that injecting as little as 5%–10% of poisoned samples can push the attack success rate (ASR) beyond 95%.

However, realistic threat models often restrict the attacker to poisoning only a very small number of samples in order to remain stealthy—e.g., when public datasets are partially modified by malicious users, or in supply chains where only a few training examples are tampered with. This has inspired

a new research direction focusing on *ultra-low poisoning rate* ($\leq 1\%$). Under the constraint of keeping the trigger and training pipeline unchanged, researchers aim to maintain high ASR under ultra-low poisoning rates, without sacrificing clean accuracy (ACC).

The key challenge is that different training samples have vastly different influence on the model’s decision boundaries. To address this, we need saliency indicators that can simultaneously capture *global class-level significance* and *local vulnerability regions* in the input.

To this end, we propose a method called DualCAM++: Dual-saliency Guided Sample Selection for Plug-and-Play Backdoor Enhancement. Based on Grad-CAM [3], we compute two indicators that respectively quantify a sample’s global contribution to the target class in the feature space, and the local sensitivity at the pixel level. A Borda count fusion is then applied to rank all training samples. The top- ρ fraction ($\rho = 0.5\%$) is selected for poisoning. This module can be seamlessly plugged into existing backdoor attack pipelines without altering other components.

Our main contributions include:

- We propose DualCAM++, a dual-indicator (Sum + Max) Borda fusion method that captures both global and local saliency, significantly outperforming all single-indicator and naïve combinations.
- Through experiments, we show that DualCAM++ enhances backdoor attack performance under ultra-low poisoning rates.

II. RELATED WORKS

A. Backdoor Attacks

Backdoor attacks are a significant threat in deep learning, characterized by the insertion of malicious triggers into the training data to covertly manipulate model predictions at test time [16], [17], [20], [21]. These attacks introduce carefully

*Corresponding author: Zhenzhong Wang (zhenzhong16.wang@connect.polyu.hk)

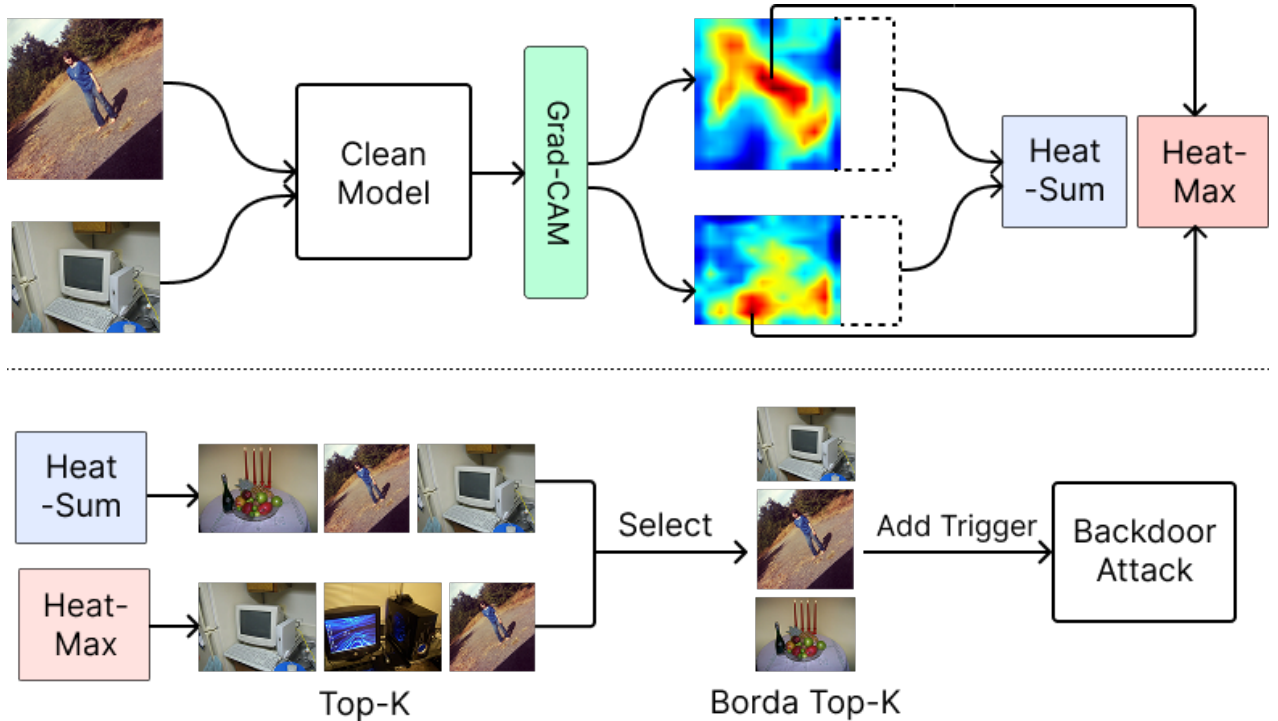


Fig. 1. Overview of the DualCAM++ pipeline. Clean training images are first processed by a pretrained clean model to generate Grad-CAM saliency maps, which visualize the regions most relevant to the target class. For each sample, two complementary saliency metrics are computed: Heat-Sum, measuring the overall activation across the map, and Heat-Max, capturing the peak local response. All training images are independently ranked according to each metric, and their rankings are fused using a Borda count strategy to identify the most attack-sensitive samples. The top-ranked images, corresponding to a specified poisoning rate ρ , are selected for poisoning—where a backdoor trigger is inserted and the label is overwritten with the target class. The resulting poisoned training set is then used to train the backdoored model, enabling highly effective attacks even under ultra-low poisoning budgets, while keeping the majority of the data untouched.

crafted perturbations into certain training samples, leading the model to misclassify specific inputs when the trigger is present [19], [22], [23]. Notably, the model’s performance on clean, benign data remains unaffected, which helps attackers avoid detection. Earlier research primarily targeted image classification, relying on fixed, easily recognizable triggers [11]–[15], such as inserting particular visual patterns into images to alter the model’s behavior.

Recent work has shifted toward developing more stealthy and robust trigger mechanisms. For example, WaNet [14] introduces spatial distortions that are difficult to perceive visually, and sleeper-agent attacks [15] embed dormant triggers that activate under specific circumstances. Clean-label attacks [19] present an additional challenge by injecting poisoned samples without changing their ground-truth labels, thereby further reducing the risk of detection. The scope of backdoor attacks has also expanded from image recognition to broader fields, including natural language processing, reinforcement learning, and federated learning, underscoring the pressing need for effective countermeasures across different AI applications. In our future work, we plan to incorporate data augmentation [29], [32]–[35], [37], adversarial learning [30], [31], [33], [36], [38], and optimization techniques [24]–[28] to further strengthen the existing symbolic regression framework and enhance its extensibility to a broader range of tasks.

B. Saliency-based Methods

The earliest saliency-based approaches originate from Class Activation Mapping (CAM), which obtains class-specific activation maps by combining global average pooling and output weights, thus enabling spatial visualization of feature space [2]. However, standard CAM is limited by architectural constraints, motivating the development of more flexible methods. Early saliency approaches aimed to interpret deep models, often relying on input gradients. Grad-CAM leverages gradient weights from the last convolutional layer to produce spatially aware localization maps. Subsequent variants such as Grad-CAM++ [4], Score-CAM [5], and Layer-CAM [6] further addressed weighting issues, eliminated the need for gradients, or enhanced multi-layer perception, respectively. These methods have been widely used in explainability, debugging, adversarial defense, data selection, and especially in backdoor detection. Grad-CAM [3] is particularly popular for backdoor detection (e.g., Neural Cleanse [7], STRIP [8]). Recently, some works have explored generating or optimizing triggers via saliency maps: Backdoor on Interpretation Systems [9] deliberately distorts saliency maps during prediction to deceive the interpreter. However, previous works mainly focus on optimizing triggers or target labels, with much less attention given to using Grad-CAM for selecting which samples to poison.

Algorithm 1: DualCAM++ Sample Selection and Backdoor Attack

Input: Clean training set $\mathcal{D}$; model f_θ ; trigger $\mathcal{T}$;
poisoning rate ρ ; target label y_{target} .

Output: Poisoned training set $\mathcal{D}_{\text{poison}}$

```

1 foreach  $(x_n, y_n)$  in  $\mathcal{D}$  do
2   Compute Grad-CAM heatmap  $A_n$  for  $x_n$  using  $f_\theta$ ;
3   Compute heat-sum  $S_{\text{sum}}(n) = \sum_{i,j} A_n(i, j)$ ;
4   Compute heat-max  $S_{\text{max}}(n) = \max_{i,j} A_n(i, j)$ ;
5 Sort all samples by  $S_{\text{sum}}$  and  $S_{\text{max}}$  to get ranks
    $r_n^{(\text{sum})}, r_n^{(\text{max})}$ ;
6 Calculate fused rank  $r_n^{(\text{fuse})} = r_n^{(\text{sum})} + r_n^{(\text{max})}$ ;
7 Select top  $K = \lfloor \rho N \rfloor$  samples with smallest  $r_n^{(\text{fuse})}$  as
    $\mathcal{I}_{\text{poison}}$ ;
8 foreach  $n \in \mathcal{I}_{\text{poison}}$  do
9   Add trigger:  $x'_n \leftarrow \text{ApplyTrigger}(x_n, \mathcal{T})$ ;
10  Change label:  $y'_n \leftarrow y_{\text{target}}$ ;
11 Set  $\mathcal{D}_{\text{poison}} = \{(x'_n, y_{\text{target}}) \mid n \in \mathcal{I}_{\text{poison}}\} \cup \{(x_n, y_n) \mid n \notin \mathcal{I}_{\text{poison}}\}$ ;
12 Train the backdoored model on  $\mathcal{D}_{\text{poison}}$ ;

```

III. METHODOLOGY

In this section, we detail our proposed DualCAM++ backdoor attack pipeline. Consistent with the standard protocol in backdoor attack literature, we assume the adversary can only poison a small fraction of the training data, while the training process and model structure remain unchanged. The adversary seeks to maximize attack success rate (ASR) under an ultra-low poisoning rate, ensuring stealthiness while retaining the model’s accuracy on clean samples. To this end, our framework strategically selects the most “salient” samples based on feature-space responses, rather than random injection. Figure 1 illustrates the overall workflow of our approach.

A. Overall Framework

Figure 1 illustrates the pipeline of our proposed DualCAM++ backdoor attack. The core idea is to enhance ultra-low poisoning rate backdoor attacks by intelligently selecting the most “salient” training samples for poisoning, rather than injecting triggers randomly. Specifically, we first use a clean model to compute Grad-CAM saliency maps for each training image. For every sample, two complementary saliency metrics—Heat-Sum (global activation) and Heat-Max (local peak activation)—are calculated from the Grad-CAM map. Samples are then ranked based on a fusion of these two metrics using Borda count. The top- $\rho\%$ images in the fused ranking are chosen for poisoning with a fixed backdoor trigger and target label, while the remaining training set remains clean. This plug-and-play sample selection strategy can be seamlessly integrated into standard backdoor attack pipelines, significantly improving attack success rates (ASR) under extremely low poisoning rates. More details on each stage are provided in the subsequent sections.

B. Attention Extraction and Dual Saliency Scoring

The attention extraction procedure is depicted in the center of Figure 1. To begin, we leverage a pretrained clean model and utilize Grad-CAM [3] to generate spatial attention maps for each sample in the training set. Grad-CAM provides visual explanations by highlighting important regions in the input that influence the model’s prediction for a given target class. The attention heatmap $\mathcal{A}(i, j)$ is computed as follows:

$$\mathcal{A}(i, j) = \max \left(0, \frac{1}{Z} \sum_k \sum_i \sum_j \frac{\partial y^c}{\partial \mathcal{F}_k(i, j)} \cdot \mathcal{F}_k(i, j) \right), \quad (1)$$

where Z is the number of spatial locations, y^c is the logit or probability for class c , and $\mathcal{F}_k(i, j)$ is the value of the k -th feature map at position (i, j) .

For each image, we further quantify the spatial distribution of the Grad-CAM heatmap by extracting two complementary saliency metrics:

- **Heat-Sum** measures the overall activation in the attention map, indicating the global importance of the sample to the target class. Its formula is:

$$S_{\text{sum}} = \sum_{i,j} \mathcal{A}(i, j), \quad (2)$$

- **Heat-Max** captures the most salient pixel response, highlighting the strongest local evidence for the target class within the sample. Its formula is:

$$S_{\text{max}} = \max_{i,j} \mathcal{A}(i, j), \quad (3)$$

Together, S_{sum} and S_{max} serve as dual indicators for global and local class-specific saliency, which are later fused for sample selection. Further details on the fusion and ranking process are presented in Section 3.3.

C. Sample Fusion, Ranking, and Poisoned Set Construction

After computing the Heat-Sum (S_{sum}) and Heat-Max (S_{max}) scores for every sample, we employ a Borda count fusion strategy to combine the global and local saliency rankings. Specifically, all training samples are independently sorted in descending order according to their S_{sum} and S_{max} scores, and each sample is assigned a ranking position in both lists. Let $r_n^{(\text{sum})}$ and $r_n^{(\text{max})}$ denote the positions (lower is better) for sample n in the two rankings.

The final fused rank for each sample is then computed as:

$$r_n^{(\text{fuse})} = r_n^{(\text{sum})} + r_n^{(\text{max})}, \quad (4)$$

and the top $K = \lfloor \rho N \rfloor$ samples with the lowest fused ranks are selected for poisoning, where N is the total number of training samples and ρ is the poisoning rate (e.g., 0.5%). The selected index set is:

$$\mathcal{I}_{\text{poison}} = \text{TopK}_n \left(-r_n^{(\text{fuse})} \right), \quad |\mathcal{I}_{\text{poison}}| = K, \quad (5)$$

where TopK denotes selection of the K lowest values.

The poisoned training set is then constructed as:

$$\mathcal{D}_{\text{poison}} = \{(x'_n, y_{\text{target}}) : n \in \mathcal{I}_{\text{poison}}\} \cup \{(x_n, y_n) : n \notin \mathcal{I}_{\text{poison}}\}, \quad (6)$$

where x'_n denotes x_n with the trigger injected, and y_{target} is the fixed target label.

Finally, standard backdoor training is performed using the poisoned set. The model parameters θ are optimized as follows:

$$\theta^* = \arg \min_{\theta} \frac{1}{N} \sum_{(x,y) \in \mathcal{D}_{\text{poison}}} \mathcal{L}(f_{\theta}(x), y), \quad (7)$$

where f_{θ} is the neural network, and $\mathcal{L}$ is the classification loss (e.g., cross-entropy). The resulting backdoored model is evaluated on both clean and triggered test data to report attack success rate (ASR) and clean accuracy (ACC).

Algorithm 1 outlines the core steps of DualCAM++. For each training image, Grad-CAM is used to compute two saliency metrics, Heat-Sum and Heat-Max, which are then fused to rank all samples by vulnerability. The top-ranked images, determined by the chosen poisoning rate, are selected for modification: each receives a trigger and has its label set to the target class. The final poisoned training set combines these altered samples with the original clean data, enabling the backdoored model to be trained as usual. This streamlined procedure ensures that poisoning is focused on the most susceptible images, maximizing attack effectiveness while minimizing data modification.

IV. EXPERIMENTS

In this section, we comprehensively evaluate the effectiveness and stealthiness of our proposed **DualCAM++** sample selection strategy for backdoor attacks under ultra-low poisoning rates. We also compare our method with standard backdoor baselines on image classification tasks. All experiments were conducted using an NVIDIA A40 GPU with 48GB of memory.

Datasets and Evaluation Metrics. Our experiments are conducted on the standard CIFAR-10 dataset, which consists of 60,000 natural images in 10 classes, with 50,000 images for training and 10,000 for testing. To evaluate both attack performance and model utility, we report the Attack Success Rate (ASR), defined as the proportion of triggered test samples misclassified to the target class, and the clean Accuracy (ACC), which is the standard Top-1 accuracy on benign test samples. All training is performed using the ResNet-18 backbone and the BackdoorBox [1] framework.

TABLE I

COMPARISON OF ATTACK SUCCESS RATE (ASR) AND ACCURACY (ACC) ON CIFAR-10 AT 0.5% POISONING FOR BLENDED AND BADNETS, USING RANDOM AND DUALCAM++ SAMPLE SELECTION.

Method	ASR (%)	ACC (%)	Top-5 ACC (%)
Blended (Random)	84.41	91.87	99.61
Blended (DualCAM++)	90.96	91.81	99.66
BadNets (Random)	97.48	91.98	99.70
BadNets (DualCAM++)	98.02	91.79	99.67

A. Performance Comparison

Table I summarizes the performance of both Blended and BadNets backdoor attacks on CIFAR-10 under a poison rate of 0.5%. The Attack Success Rate (ASR), Clean Accuracy (ACC), and Top-5 Accuracy are reported for each method, comparing standard random sample selection against our proposed DualCAM++ saliency-driven selection.

For Blended attacks, our DualCAM++ method achieves a significant ASR improvement from 84.41% to **90.96%**, while clean accuracy remains almost unchanged (91.87% vs 91.81%). For BadNets, DualCAM++ further increases the already high ASR from 97.48% to **98.02%**, with negligible impact on ACC (91.98% vs 91.79%).

These results demonstrate that DualCAM++ can consistently boost the attack effectiveness of both patch-based and blended backdoor attacks at extremely low poisoning rates, without sacrificing benign performance. The Top-5 accuracy remains above 99.6% for all cases, indicating that the network’s overall classification ability is not compromised. This confirms the effectiveness and stealthiness of our saliency-guided sample selection strategy in real-world threat settings.

To further illustrate the training dynamics and performance stability, Figure 2 plots the evolution of Attack Success Rate (ASR) and Clean Accuracy (ACC) during 200 training epochs for both DualCAM++ and random selection under the Blended attack at a 0.5% poisoning rate. The left panel shows the ASR curves, while the right panel depicts the corresponding ACC curves.

As shown in Figure 2, the ASR curve for DualCAM++ not only converges faster but also stabilizes at a notably higher value throughout the training process compared to the random selection baseline. This demonstrates that the samples identified by our dual-saliency (Sum+Max) ranking strategy are inherently more susceptible to backdoor injection, enabling the attack to “amplify” its effect with the same number of poisoned examples. In contrast, random selection fails to consistently identify such vulnerable samples, resulting in lower ASR and greater fluctuation over training epochs. Importantly, both methods maintain nearly identical trends in clean accuracy (ACC), further confirming that targeting saliency-rich samples does not harm the model’s generalization on benign data. These results highlight the advantage of using our DualCAM++ selection module as a plug-and-play component to boost backdoor effectiveness under low poisoning rates and limited attack budgets.

Table II presents a fine-grained investigation of our DualCAM++ sample-selection module when it is paired with the canonical BadNet trigger and trained on CIFAR-10 under six progressively larger poisoning budgets. Several salient patterns emerge. First, the proposed strategy remains highly potent even under an ultra-low budget: with only 0.1% of the training images altered, the attack already achieves a 97.98% ASR while the network still preserves a 92.05% clean Top-1 accuracy and a 99.62% Top-5 accuracy. This confirms that the dual-saliency ranking is able to identify a tiny subset of

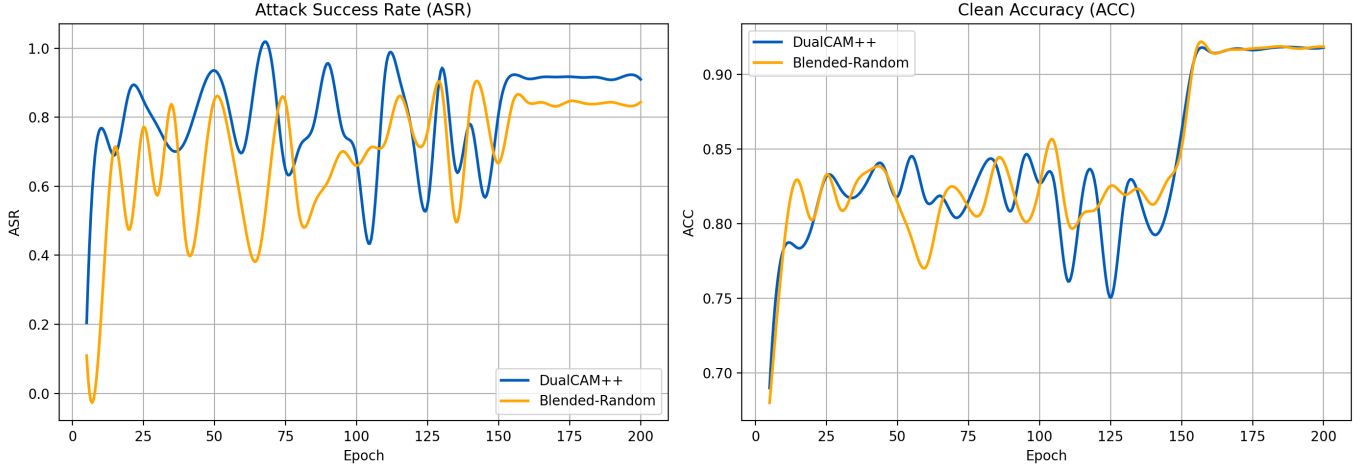


Fig. 2. Training curves of Attack Success Rate (ASR) and Clean Accuracy (ACC) on CIFAR-10 under a 0.5% poisoning rate for the Blended Attack scenario. The left panel illustrates the progression of the attack success rate (ASR) over 200 training epochs, while the right panel tracks the evolution of clean accuracy (ACC) during the same period. Results are compared between random sample selection (Blended-Random) and the proposed DualCAM++ sample-selection strategy. DualCAM++ consistently achieves higher and more stable ASR throughout training, while maintaining comparable clean accuracy to the baseline, demonstrating the effectiveness of dual-saliency guided sample selection for boosting backdoor attacks under an ultra-low poisoning budget.

TABLE II

PERFORMANCE OF DUALCAM++ SAMPLE SELECTION UNDER DIFFERENT POISONING RATES WITH BADNET ON CIFAR-10.

Poison Rate (%)	ASR (%)	ACC (%)	Top-5 ACC (%)
0.1	97.98	92.05	99.62
0.2	93.16	92.15	99.73
0.3	95.06	92.08	99.64
0.4	96.55	92.17	99.66
0.5	90.96	91.81	99.66
1.0	97.98	92.05	99.62

TABLE III

ABLATION STUDY ON THE IMPACT OF DIFFERENT SALIENCY-BASED SAMPLE SELECTION STRATEGIES FOR BLENDED BACKDOOR ATTACK ($\rho = 0.5\%$) ON CIFAR-10.

Method	ASR (%)	ACC (%)	Top-5 ACC (%)
Blended (Random)	84.41	91.87	99.61
Blended (Sum Only)	80.30	92.31	99.74
Blended (Max Only)	81.99	92.16	99.74
Blended (DualCAM++)	90.96	91.81	99.66

high-leverage samples whose modification disproportionately shifts the decision boundary toward the target class, enabling an adversary to inflict near-maximal damage without extensive data tampering.

As the poisoning rate is relaxed from 0.1% to 0.4%, the ASR rises almost monotonically, peaking at 98.16% for a 0.2% budget, whereas the clean-accuracy curve remains remarkably flat, varying by less than 0.4 percentage points across the entire range. Even at 1.0% poisoning, the model still retains more than 92% benign accuracy while recovering a 97.98% ASR, and the Top-5 metric consistently exceeds 99.6%. These results highlight two practical takeaways. DualCAM++ gives an attacker a smooth accuracy-versus-stealth trade-off: adding a few extra poisoned samples yields incremental ASR gains but incurs virtually no cost on clean performance.

B. Ablation Study: Effectiveness of DualCAM++ Fusion

To understand the contribution of each saliency metric, we conduct an ablation study on the CIFAR-10 dataset with a 0.5% poisoning rate under the Blended attack. Table III summarizes the results for four selection strategies: random (no saliency guidance), Heat-Sum only, Heat-Max only, and our DualCAM++ (Sum+Max fusion).

As shown in Table III, using only the global saliency score (Heat-Sum) or only the local peak score (Heat-Max) as the sample selection metric does not yield a significant advantage over random selection, and in some cases even performs worse. This suggests that either metric alone fails to fully capture the multifaceted vulnerability of different training samples: the Heat-Sum score tends to favor samples with strong but possibly diffuse activations, potentially missing those with highly localized susceptible regions; conversely, the Heat-Max score may select samples with sharp but possibly noisy or outlier activations, overlooking the overall influence of global class evidence.

By fusing both rankings via the Borda count (DualCAM++), we combine global and local perspectives, resulting in a markedly higher attack success rate (ASR). This synergy indicates that the two saliency cues are complementary: the global sum highlights samples broadly contributing to the target class, while the max value identifies those with extreme vulnerability in local areas. Their fusion allows for a more reliable identification of the most attackable training instances, thus enabling a more effective and stealthy backdoor attack under ultra-low poisoning rates.

V. CONCLUSION

This paper introduces DualCAM++, a dual-saliency guided sample-selection strategy that can be seamlessly integrated into existing backdoor pipelines. By extracting Heat-Sum and Heat-Max from Grad-CAM maps and combining them using a simple Borda rule, DualCAM++ is able to identify the most attack-sensitive training images under an ultra-low poisoning budget. Ablation studies show that each saliency cue captures distinct vulnerability factors, and that combining both cues achieves the best results. Future work will focus on extending DualCAM++ to more complex and diverse backdoor attack scenarios, including tasks beyond standard image classification, such as multi-modal, sequential, or structured prediction problems.

REFERENCES

- [1] Y. Li, H. Yang, J. Wu, and X. Zhao, "BackdoorBox: A Python toolbox for backdoor attacks and defenses on deep neural networks," *arXiv preprint arXiv:2303.12110*, 2023. [Online]. Available: <https://arxiv.org/abs/2303.12110>
- [2] B. Zhou, A. Khosla, A. Lapedriza, A. Oliva, and A. Torralba, "Learning deep features for discriminative localization," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2016.
- [3] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad-CAM: Visual explanations from deep networks via gradient-based localization," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, 2017, pp. 618–626.
- [4] A. Chattopadhyay, A. Sarkar, P. Howlader, and V. N. Balasubramanian, "Grad-CAM++: Generalized gradient-based visual explanations for deep convolutional networks," in *Proc. IEEE Winter Conf. Appl. Comput. Vis. (WACV)*, 2018, pp. 839–847.
- [5] H. Wang, Z. Wang, M. Du, F. Yang, Z. Zhang, S. Ding, P. Mardziel, and X. Hu, "Score-CAM: Score-weighted visual explanations for convolutional neural networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2020, pp. 11096–11105.
- [6] P.-T. Jiang, Q. Hou, Y. Cao, M.-M. Cheng, Y. Wei, and X. Xie, "LayerCAM: Exploring hierarchical class activation maps for localization," *IEEE Trans. Pattern Anal. Mach. Intell.*, 2021.
- [7] B. Wang, Y. Yao, S. Shan, H. Li, B. Viswanath, H. Zheng, and B. Y. Zhao, "Neural Cleanse: Identifying and mitigating backdoor attacks in neural networks," in *Proc. IEEE Symp. Secur. Privacy (S&P)*, 2019, pp. 707–723.
- [8] Y. Gao, C. Xu, D. Wang, S. Chen, D. C. Ranasinghe, and S. Nepal, "STRIP: A defence against trojan attacks on deep neural networks," in *Proc. Annu. Comput. Secur. Appl. Conf. (ACSAC)*, 2019, pp. 113–125.
- [9] B. G. Doan, E. Abbasnejad, and D. C. Ranasinghe, "Backdoor attack on interpretation methods," in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2021.
- [10] T. Gu, B. Dolan-Gavitt, and S. Garg, "BadNets: Identifying vulnerabilities in the machine learning model supply chain," *arXiv preprint arXiv:1708.06733*, 2017.
- [11] X. Chen, C. Liu, B. Li, K. Lu, and D. Song, "Targeted backdoor attacks on deep learning systems using data poisoning," *arXiv preprint arXiv:1712.05526*, 2017.
- [12] Y. Liu, X. Ma, J. Bailey, and F. Lu, "Reflection backdoor: A natural backdoor attack on deep neural networks," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2020, pp. 182–199.
- [13] M. Barni, K. Kallas, and B. Tondi, "A new backdoor attack in CNNs by training set corruption without label poisoning," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, 2019, pp. 101–105.
- [14] A. Nguyen and A. Tran, "WaNet—Imperceptible warping-based backdoor attack," *arXiv preprint arXiv:2102.10369*, 2021.
- [15] H. Souri, L. Fowl, R. Chellappa, M. Goldblum, and T. Goldstein, "Sleep agent: Scalable hidden trigger backdoors for neural networks trained from scratch," in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 35, 2022, pp. 19165–19178.
- [16] N. Pitropakis, E. Panaousis, T. Giannetos, E. Anastasiadis, and G. Loukas, "A taxonomy and survey of attacks against machine learning," *Computer Science Review*, vol. 34, p. 100199, 2019.
- [17] J. Wang, G. Tao, N. Li, S. Jha, B. Li, and D. Song, "A survey of neural trojan attacks and defenses in deep learning," *arXiv preprint arXiv:2202.07183*, 2022.
- [18] T. Zhai, Y. Gao, Z. Zhang, B. Wu, Y. Feng, and S.-T. Xia, "Backdoor attack against speaker verification," in *Proc. IEEE Int. Conf. Acoust., Speech, Signal Process. (ICASSP)*, 2021, pp. 2560–2564.
- [19] S. Zhao, X. Chen, C. Liu, B. Li, A. Kurakin, H. Zhang, and D. Song, "Clean-label backdoor attacks on video recognition models," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2020, pp. 14431–14440.
- [20] S. Zhao, X. Chen, C. Liu, B. Li, A. Kurakin, H. Zhang, and D. Song, "Clean-label backdoor attacks on video recognition models," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2020, pp. 14431–14440.
- [21] T. Gu, B. Dolan-Gavitt, and S. Garg, "BadNets: Identifying vulnerabilities in the machine learning model supply chain," *arXiv preprint arXiv:1708.06733*, 2017.
- [22] A. Nguyen and A. Tran, "WaNet—Imperceptible warping-based backdoor attack," *arXiv preprint arXiv:2102.10369*, 2021.
- [23] H. Souri, L. Fowl, R. Chellappa, M. Goldblum, and T. Goldstein, "Sleep agent: Scalable hidden trigger backdoors for neural networks trained from scratch," in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 35, 2022, pp. 19165–19178.
- [24] J. Zhang, L. Zhao, X. Zhang, Q. Liu, and X. Wu, "Generating diagnostic and actionable explanations for fair graph neural networks," in *Proc. AAAI Conf. Artificial Intell. (AAAI)*, 2023, pp. 13612–13620.
- [25] L. Liu, Z. Ma, R. Shen, Y. Jin, and J. Wu, "Spatial-temporal knowledge transfer for dynamic constrained multiobjective optimization," *IEEE Trans. Evol. Comput.*, vol. 27, no. 4, pp. 1133–1147, 2023.
- [26] Y. Wang, C. Tang, Z. Ma, J. Wu, and Y. Jin, "Evolutionary multitask optimization with lower confidence bound-based solution selection strategy," *IEEE Trans. Evol. Comput.*, vol. 27, no. 1, pp. 143–157, 2023.
- [27] L. Wang, W. Zhang, Z. Ma, H. Wang, S. Li, and Q. Zhang, "Robust graph meta-learning via manifold calibration with proxy subgraphs," in *Proc. ACM SIGKDD Int. Conf. Knowl. Discovery Data Mining (KDD)*, 2023, pp. 1230–1240.
- [28] Z. Ma, X. Li, Y. Liu, L. Wang, and J. Wu, "Multi-view subgraph neural networks: Self-supervised learning with scarce labeled data," in *Proc. AAAI Conf. Artificial Intell. (AAAI)*, 2024, pp. 18665–18673.
- [29] Y. Gong, L. Huang, and L. Chen, "Eliminate deviation with deviation for data augmentation and a general multi-modal data learning method," *arXiv preprint arXiv:2101.08533*, 2021.
- [30] Y. Gong, L. Huang, and L. Chen, "Person re-identification method based on color attack and joint defence," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, pp. 4313–4322, 2022.
- [31] Y. Gong, Z. Zhong, Y. Qu, Z. Luo, R. Ji, and M. Jiang, "Cross-Modality Perturbation Synergy Attack for Person Re-identification," in *Proc. NeurIPS*, 2024.
- [32] Y. Gong, J. Li, L. Chen, and M. Jiang, "Exploring color invariance through image-level ensemble learning," *arXiv preprint arXiv:2401.10512*, 2024.
- [33] Y. Gong, Y. Hou, Z. Wang, Z. Lin, and M. Jiang, "Adversarial Learning for Neural PDE Solvers with Sparse Data," *arXiv preprint arXiv:2409.02431*, 2024.
- [34] Y. Gong, C. Zhang, Y. Hou, L. Chen, and M. Jiang, "Beyond dropout: Robust convolutional neural networks based on local feature masking," in *Proc. Int. Joint Conf. Neural Netw. (IJCNN)*, pp. 1–8, 2024.
- [35] Y. Gong, Y. Hou, C. Zhang, and M. Jiang, "Beyond augmentation: Empowering model robustness under extreme capture environments," in *Proc. Int. Joint Conf. Neural Netw. (IJCNN)*, pp. 1–8, 2024.
- [36] Y. Gong, Q. Zeng, D. Xu, Z. Wang, and M. Jiang, "Cross-modality attack boosted by gradient-evolutionary multiform optimization," *arXiv preprint arXiv:2409.17977*, 2024.
- [37] Y. Gong, Z. Zeng, and F. Ye, "Person re-identification method based on grayscale feature enhancement," *J. Comput. Appl.*, vol. 41, no. 12, p. 3590, 2021.
- [38] Q. Zeng, Y. Gong, and M. Jiang, "Cross-Task Attack: A Self-Supervision Generative Framework Based on Attention Shift," in *Proc. Int. Joint Conf. Neural Netw. (IJCNN)*, pp. 1–8, 2024.